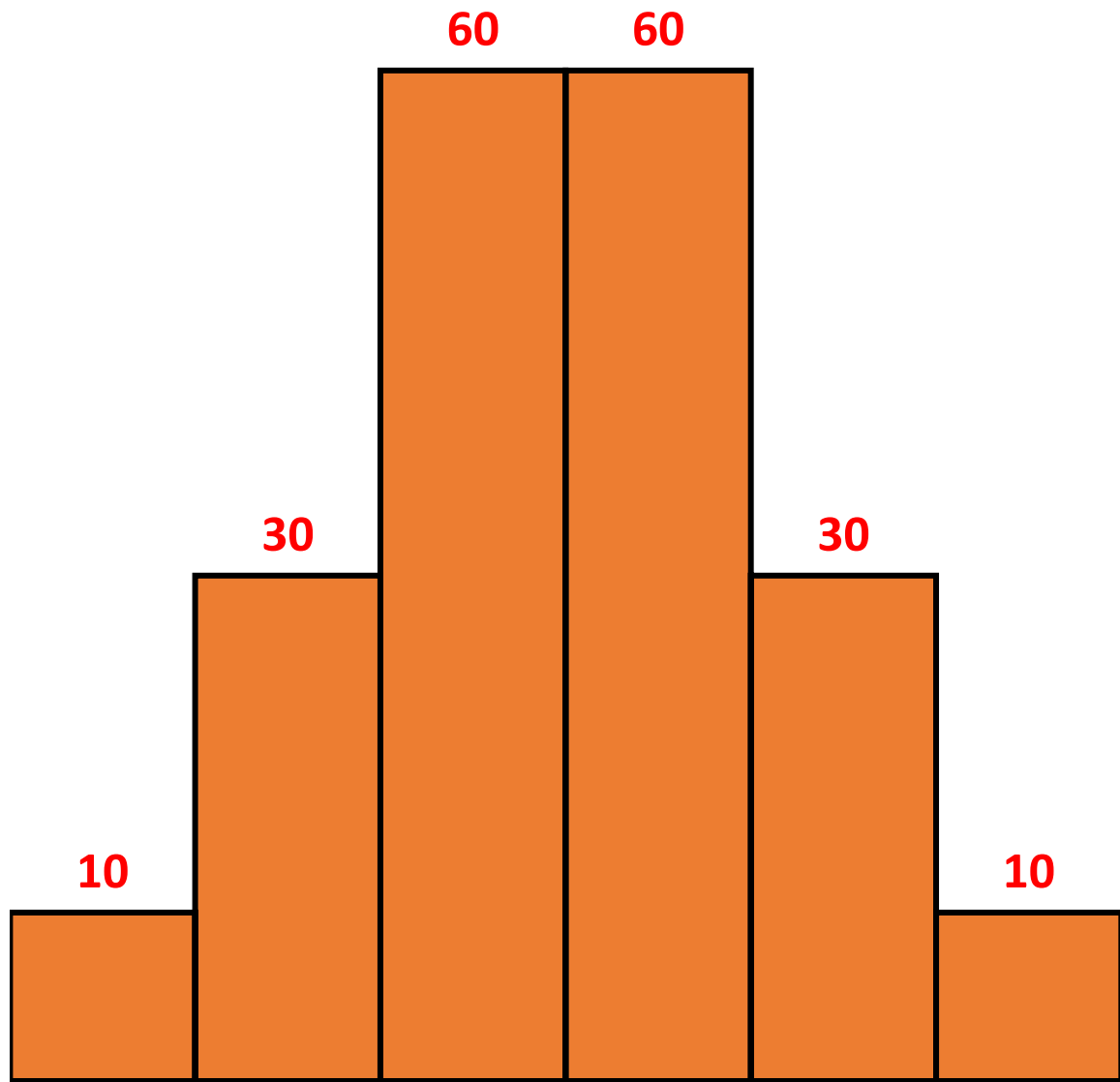
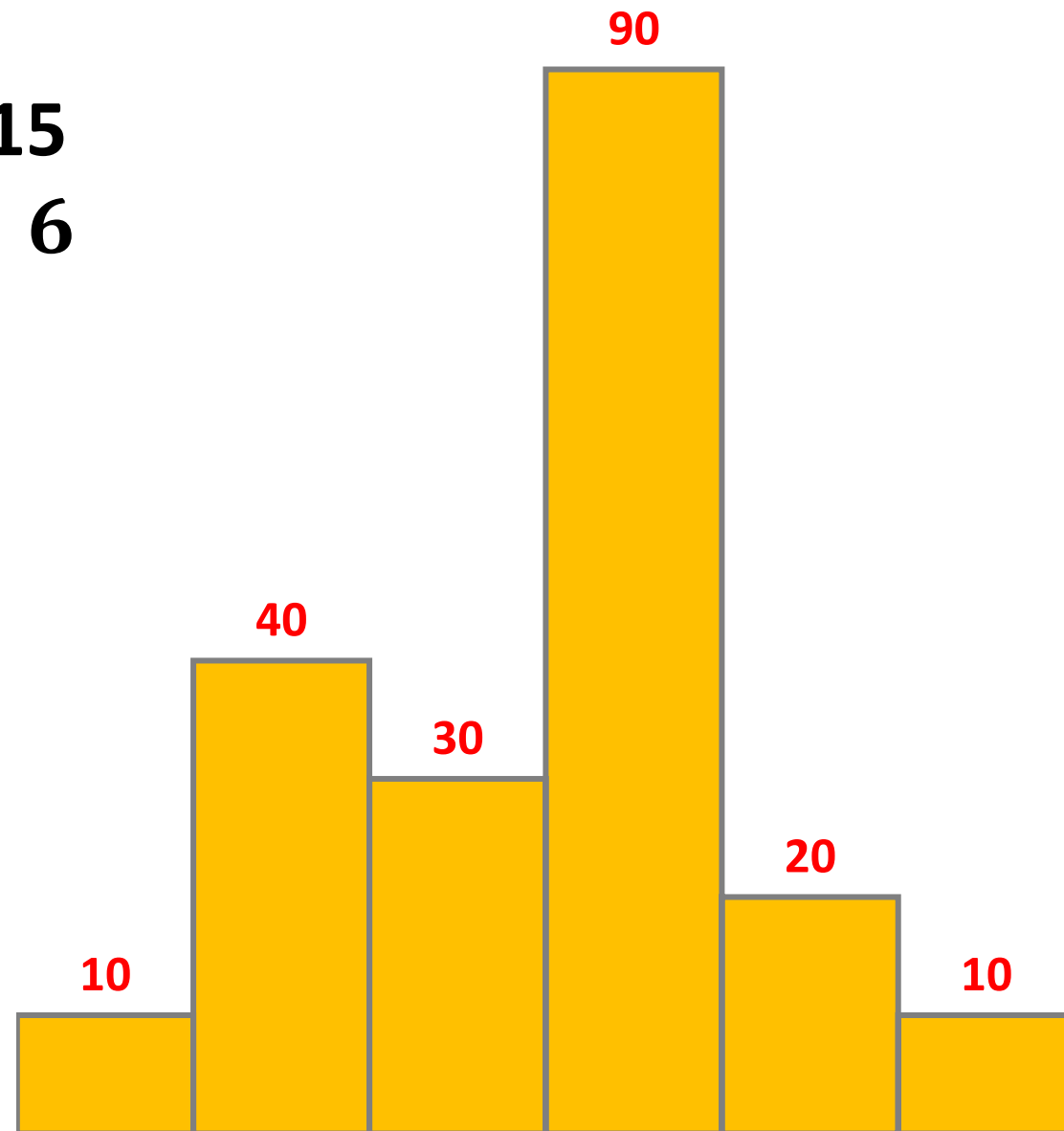




$$\bar{x} = 15$$
$$\sigma = 6$$



Skewness

When a series is not symmetrical it is said to be asymmetrical or *skewed*.

A distribution is said to be ‘skewed’ when the *mean* and *median* fall at different points in the distribution, and the balance (or centre of gravity) is shifted to one side or the other to left or right.

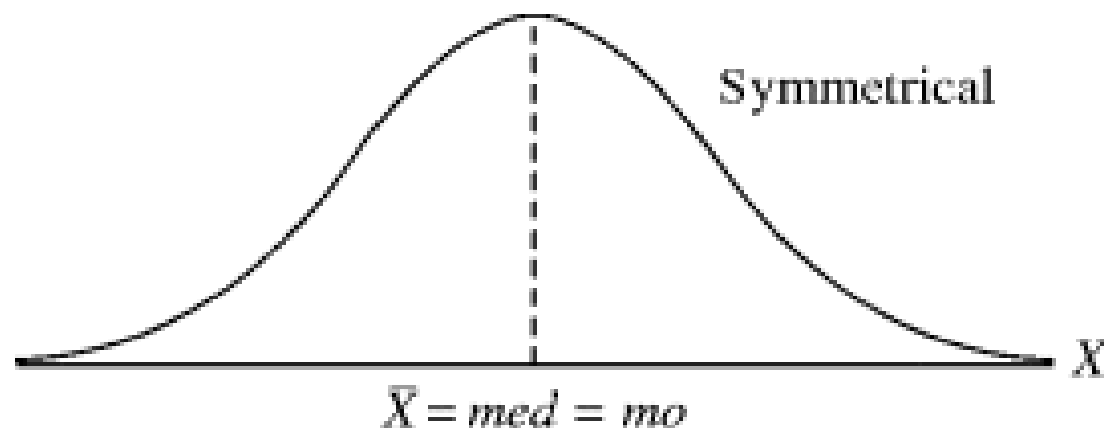
Measure of skewness is the lack of symmetry of a distribution.

Dispersion is concerned with the amount of variation rather than with its direction.

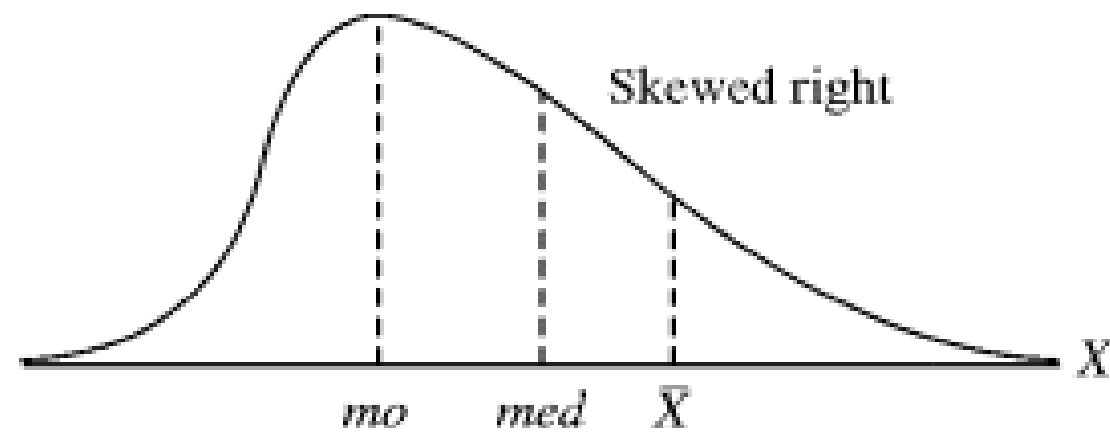
Skewness tell us about the *direction of the variation* or the departure from symmetry.

Types of Skewness:

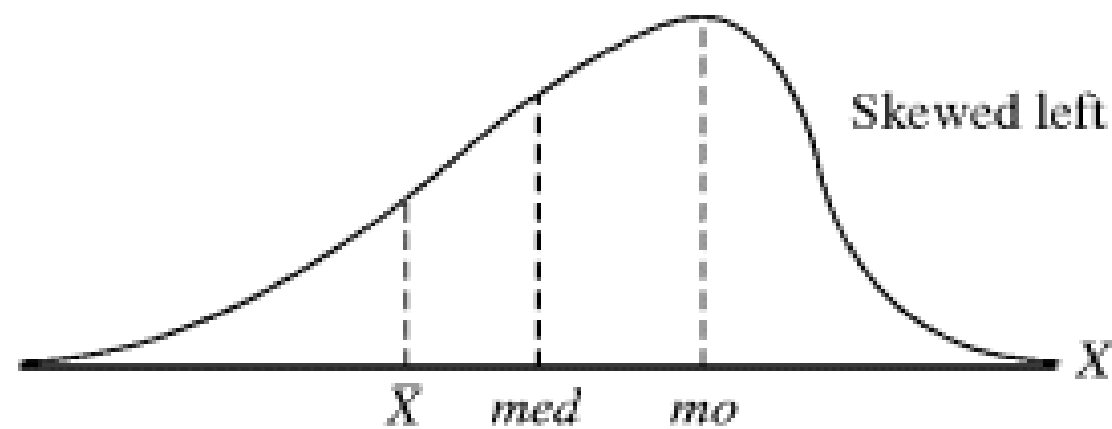
- (i) Symmetrical Distribution
- (ii) Positively Skewed Distribution (**Mean > Mode**)
- (iii) Negatively Skewed Distribution (**Mean < Mode**)



(a)



(b)



(c)

Measure of Skewness

Absolute measures of Skewness(Sk)

$$Sk = \bar{X} - \text{Mode}$$

Relative measures of Skewness:

- (i) Karl pearson's coefficient of skewness.
- (ii) Bowley's coefficient of skewness.
- (iii) Measure of skewness based on moments.

(i) Karl pearson's coefficient of skewness

$$\text{Coefficient of Skewness : } Sk = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}} = \frac{\bar{X} - Mo}{\sigma}$$

however, in practice, it is rare that the value of ***Sk*** exceed the limits of **± 1** .

But using ,

$$Mode = 3 \text{ Median} - 2 \text{ Mean}$$

$$Sk = \frac{3(\bar{X} - \text{Median})}{\sigma}$$

this measure can vary between **± 3** ;

(ii) Bowley's coefficient of skewness

It is based on Quartiles. In a symmetrical distribution first and third quartiles are equidistant from the median :



In a symmetrical distribution the third quartile is the same distance above the median as the first quartile is below it, i.e.,

$$Q_3 - \text{Median} = \text{Median} - Q_1$$

$$\text{or} \quad Q_3 + Q_1 - 2 \text{Median} = 0$$

$$\therefore Sk = \frac{Q_3 + Q_1 - 2 \text{Median}}{Q_3 - Q_1}$$

This measure is called the *quartile measure of skewness* and varies between ± 1 .

(iii) Moments

Moment is a measure of a force with respect to its tendency to provide rotation.

The strength of the tendency depends on the amount of force and the distance from the origin of the point at which the force is exerted.

Definition: Let the symbol $x_i = (X_i - \bar{X})$ be used to represent the deviation of any item in a distribution from the arithmetic average of that distribution.

The arithmetic mean of the various powers of these deviations in any distribution are called the *moments* of the distribution.

Central Moments (Moments about the Arithmetic Mean):

First Moment $\mu_1 = \frac{\sum(X_i - \bar{X})}{N}$ (sum of the deviations from A.M. is always zero. **$\mu_1 = 0$**)

Second Moment $\mu_2 = \frac{\sum(X_i - \bar{X})^2}{N} = \sigma^2 = \text{Variance}$

Third Moment $\mu_3 = \frac{\sum(X_i - \bar{X})^3}{N}$

Fourth Moment $\mu_4 = \frac{\sum(X_i - \bar{X})^4}{N}$

For a frequency distribution:

First Moment $\mu_1 = \frac{\sum f_i(X_i - \bar{X})}{N}$

Second Moment $\mu_2 = \frac{\sum f_i(X_i - \bar{X})^2}{N} = \sigma^2 = \text{Variance}$

Third Moment $\mu_3 = \frac{\sum f_i(X_i - \bar{X})^3}{N}$ or $\frac{\sum f_i x_i^3}{N}$

Fourth Moment $\mu_4 = \frac{\sum f_i(X_i - \bar{X})^4}{N}$ or $\frac{\sum f_i x_i^4}{N}$

Non-central Moments (Moments about the Assumed Mean)

Where the actual mean is in fractions it is difficult to calculate moments by applying the above formulae. In such cases we can first compute moments about an **arbitrary origin(A)**.

$$\mu'_1 = \frac{\sum(X_i - A)}{N} = \bar{X} - A \quad ; \quad \mu'_1 = \frac{\sum f_i(X_i - A)}{N}$$

$$\text{Mean} = \bar{X} = \mu'_1 + A$$

$$\mu'_2 = \frac{\sum(X_i - A)^2}{N} \quad ; \quad \mu'_2 = \frac{\sum f_i(X_i - A)^2}{N}$$

$$\mu'_3 = \frac{\sum(X_i - A)^3}{N} \quad ; \quad \mu'_3 = \frac{\sum f_i(X_i - A)^3}{N}$$

$$\mu'_4 = \frac{\sum(X_i - A)^4}{N} \quad ; \quad \mu'_4 = \frac{\sum f_i(X_i - A)^4}{N}$$

Conversion of moments about an *Arbitrary origin* into Moments about *mean*

$$\mu_1 = \mu'_1 - \mu'_1$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$

$$\beta_1 \text{ (beta one)} = \frac{\mu_3^2}{\mu_2^3} \text{ (Coeff. of Skewness)}$$

$$\beta_2 \text{ (beta two)} = \frac{\mu_4}{\mu_2^2} \text{ (Coeff. of Kurtosis)}$$

$$\gamma_1 \text{ (Gamma one)} = \sqrt{\beta_1} \text{ (Coeff. of Skewness)}$$

$$\gamma_2 \text{ (Gamma two)} = \beta_2 - 3 \text{ (Coeff. of Kurtosis)}$$

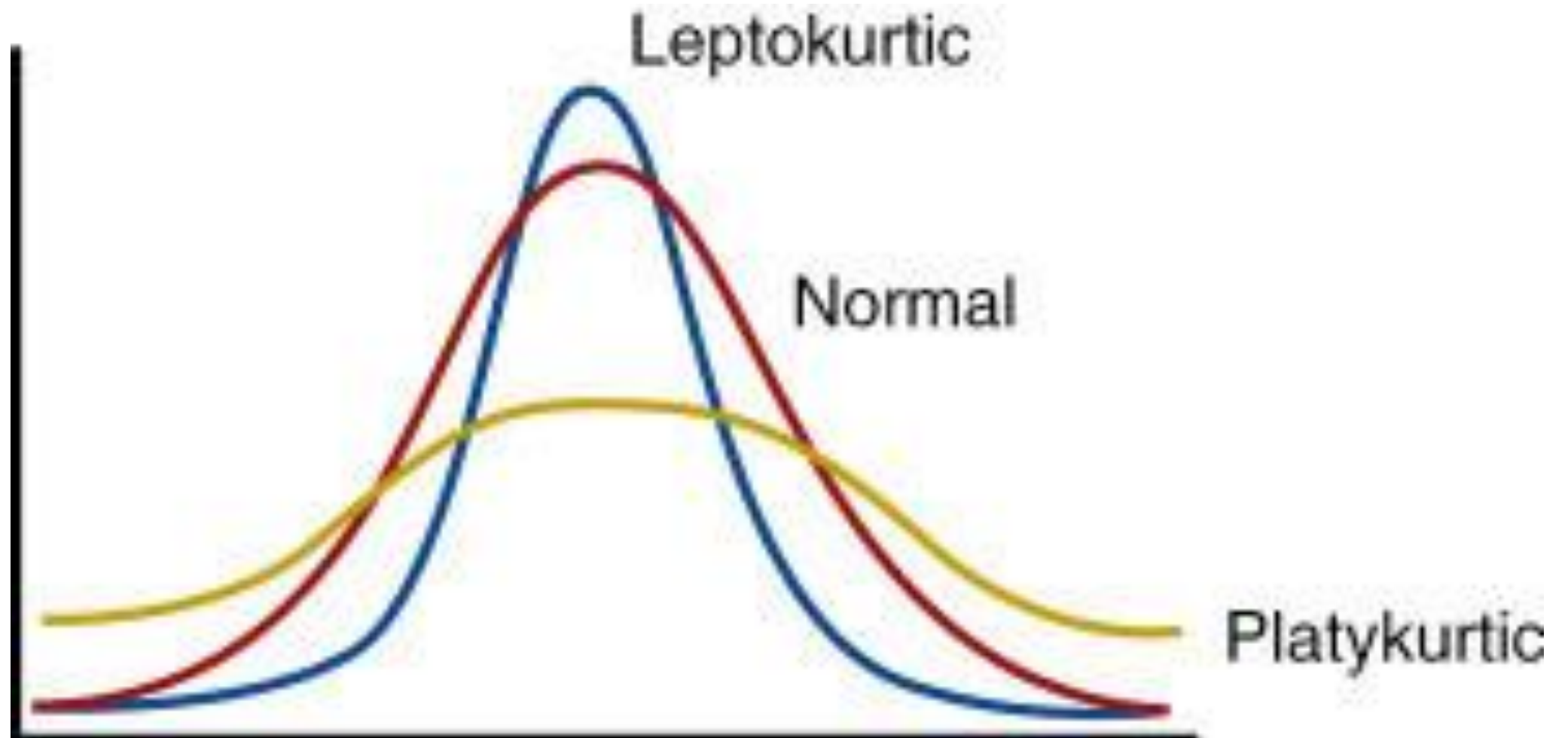
Kurtosis

Kurtosis enables us to have an idea about the ‘*flatness*’ or ‘*peakedness*’ in the region about the mode of a frequency curve.

“Convexity of the frequency curve”

It is measured by the Coefficient of $\beta_2 = \frac{\mu_4}{\mu_2^2}$ or $\gamma_2 = \beta_2 - 3$

- | | | | |
|-------|-----------------------------------|---|-------------------|
| (i) | $\beta_2 = 3, i.e., \gamma_2 = 0$ | : | mesokurtic curve |
| (ii) | $\beta_2 < 3, i.e., \gamma_2 < 0$ | : | platykurtic curve |
| (iii) | $\beta_2 > 3, i.e., \gamma_2 > 0$ | : | leptokurtic curve |



<i>Marks</i>	<i>Frequency</i>
0-10	5
10-20	20
20-30	15
30-40	45
40-50	10
50-60	5

Find the Measure of skewness based on moments.

$$\bar{X} = 30$$
$$\text{Mode} = 35.3846$$

Absolute measures of Skewness(Sk)

$$\text{Sk} = \bar{X} - \text{Mode} = -5.3846$$

<i>Marks</i>	<i>Mid point (m)</i>	<i>f</i>	<i>d = (m - 35)/10</i>	<i>fd</i>	<i>fd²</i>	<i>fd³</i>	<i>fd⁴</i>
0-10	5	5	-3	-15	45	-135	405
10-20	15	20	-2	-40	80	-160	320
20-30	25	15	-1	-15	15	-15	15
30-40	35	45	0	0	0	0	0
40-50	45	10	+1	10	10	10	10
50-60	55	5	+2	10	20	40	80
		N = 100		$\Sigma fd = -50$	$\Sigma fd^2 = 170$	$\Sigma fd^3 = -260$	$\Sigma fd^4 = 830$

$$\mu'_1 = \frac{\Sigma f_i d_i}{N} \times i = -5$$

$$\mu'_3 = \frac{\Sigma f_i d_i^3}{N} \times i^3 = 2600$$

$$\text{Mean} = \bar{X} = \mu'_1 + A = 30$$

$$\mu'_2 = \frac{\Sigma f_i d_i^2}{N} \times i^2 = 170$$

$$\mu'_4 = \frac{\Sigma f_i d_i^4}{N} \times i^4 = 83000$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = 145 = \text{variance} = \sigma^2$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3 = -300$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 = 54625$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = -0.02952 ; \quad \beta_2 = \frac{\mu_4}{\mu_2^2} = 2.5981$$

$$\gamma_1 = \sqrt{\beta_1} = -0.172 ; \quad \gamma_2 = \beta_2 - 3 = -0.4019$$

Since, γ_1 - is negative, the distribution is negatively skewed.

γ_2 - is negative, the distribution is platykurtic curve.

Example

From the following data, calculate moments about assumed mean 25 and convert them into central moments :

C.I.	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Freq.	5	12	18	40	15	7	3

State whether the distribution is leptokurtic or platykurtic.

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = 177.39 = \text{variance} = \sigma^2$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3 = 47.982$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 = 95009.364$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0.0203$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = 3.019$$

$$\gamma_1 = \sqrt{\beta_1} = 0.1424$$

$$\gamma_2 = \beta_2 - 3 = 0.019$$

Since, γ_1 - is positive, the distribution is positively skewed.

γ_2 - is negative, the distribution is leptokurtic curve.

Example

The first four moments of a distribution about value 2 are 1, 2.5, 5.5 and 16, respectively.

Calculate the first four moments about the mean and comment on the nature of the distribution.

Example

Real-Time Situation: Student Feedback on Campus Transport

Context:

You've collected student feedback scores (on a scale of 1 to 10) regarding the campus transport system. The feedback was gathered from 300 students across different hostels. You're analyzing the distribution to decide whether the transport system needs overhaul, minor tweaks, or just better communication.

Data Summary (hypothetical):

Mean score: 6.2

Standard deviation: 2.1

Skewness: -1.5

Kurtosis: 5.8

Problem Statement:

You are tasked with:

**Interpreting the skewness and kurtosis to understand student sentiment.
Identifying whether the issue is with a few extremely dissatisfied students
or a broader dissatisfaction trend.**

**Recommending whether to revise the transport schedule, improve
communication, or conduct focus group discussions.**

Skewness (−1.5):

Strong negative skew → Majority of students gave high scores, but a significant minority gave very low scores.

Indicates pockets of severe dissatisfaction, possibly due to specific routes or timings.

Kurtosis (5.8):

Leptokurtic → Distribution has a sharp peak and heavy tails.

Suggests that while most students cluster around a common opinion, there are extreme outliers (very low scores).

Implications:

The transport system is generally acceptable, but some students are facing serious issues.

These issues are likely localized (e.g., specific hostel routes, peak hour congestion).

Averages alone would mask the problem — skewness and kurtosis reveal the hidden dissatisfaction.

Suggested Action Plan:

Conduct route-specific focus group discussions to identify pain points.

Use median and mode alongside mean to report feedback more transparently.

Consider pilot schedule adjustments for routes with extreme dissatisfaction.

Communicate clearly about transport policies to reduce perception gaps.